The Spectrometer

The wavelength and refractive index can be calculated by measuring the deflection angle of light. The spectrometer is an instrument for precise measurement of deflection angle of light. We can calculate the vertex angle of a prism by measuring the reflection angle of light; calculate the refractive index of a prism by measuring the deflection angle of light; calculate the wavelength of light by measuring the grating constant.

Objectives

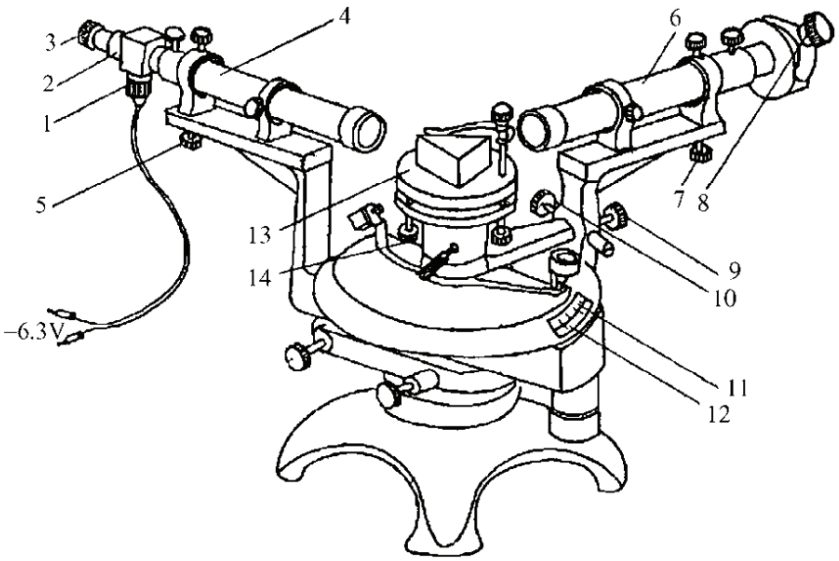
1. Learn the structure and optical principle of the spectrometer.
2. Learn the adjustment of spectrometer.
3. Measure the vertex angle of a prism by using spectrometer.

Apparatus

JJY spectrometer, prism, mirror, Sodium vapour lamp.

Theory

The structure of the spectrometer is shown in Fig. 1.



1. bulb; 2. reticle tube; 3. eyepiece; 4. telescope tube; 5. levelling screw of the telescope; 6. collimator; 7. levelling screw of the collimator; 8. slit's width adjusting screw; 9. vernier locking screw; 10. fine adjustment screw of the vernier; 11. vernier; 12. reading dial; 13. table; 14. levelling screws of the table

Fig. 1 The structure of the spectrometer

1. Telescope

The telescope is for receiving and observing light reflected or refracted by optical devices. It consists of a tube, an objective lens, an eyepiece, a reticle, a small prism, and a green bulb.

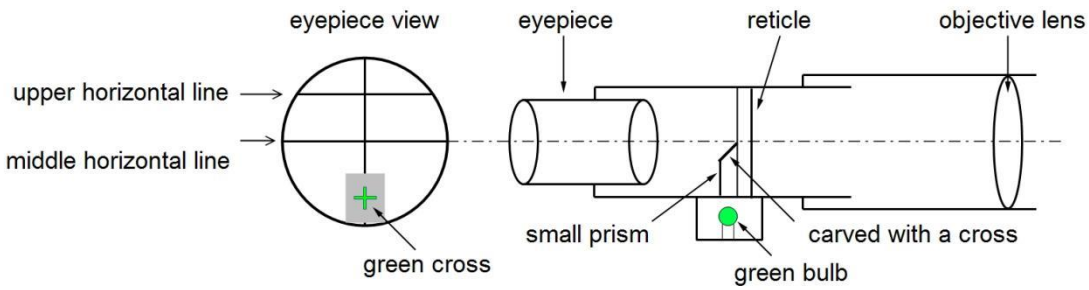


Fig. 2 The structure of the telescope

Adjust the distance between the eyepiece and the reticle, and the distance between the objective lens and the reticle, so that the reticle is on the focal plane of the eyepiece and the objective lens.

There is a small prism above the bulb, and the top surface of the small prism is carved with a cross. When the bulb is turned on, we can see a green cross and three lines in eyepiece view.

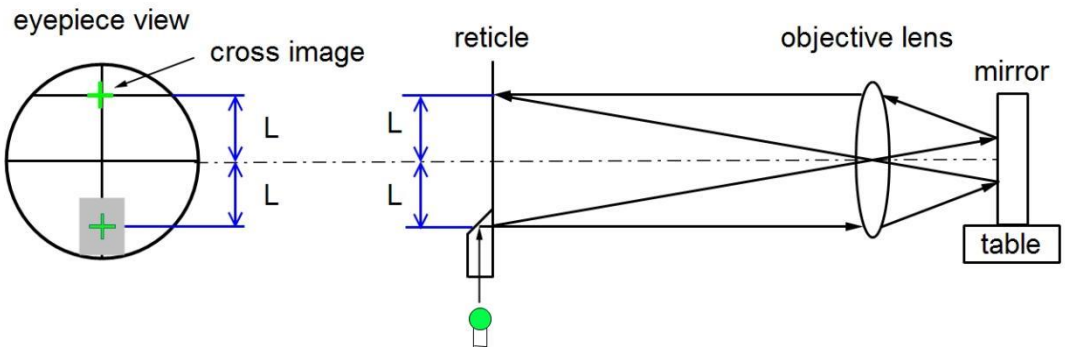


Fig. 3 Optical principle of the telescope

Place a mirror on the table, if the table is adjusted to be parallel to the horizontal plane, it means that the reflective surface of the mirror is perpendicular to the horizontal plane. When the telescope is adjusted to be parallel to the horizontal plane, the light emitted by the bulb is reflected by the small prism, through the objective lens to the mirror, and then returned to the reticle, forming a bright cross image on the upper horizontal line in eyepiece view. By observing whether the reflection cross image coincides with the upper horizontal line, it can be determined whether the telescope and table have been adjusted to be horizontal. This is the theoretical basis for adjusting the telescope and table.

2. Collimator

The collimator can produce a parallel beam of light. It consists of a tube, a lens and a slit. When the distance between the slit and the lens is adjusted to the focus of the lens, a parallel beam of light can be emitted by the collimator.

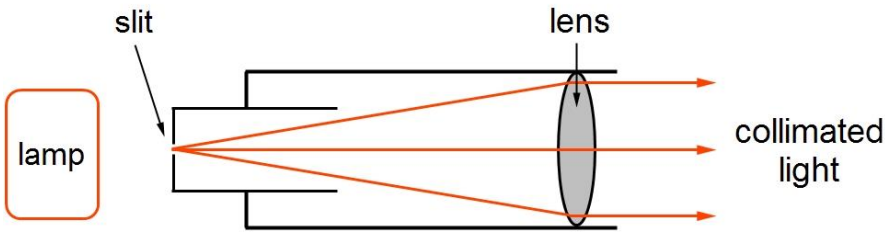


Fig. 4 Optical principle of the collimator

3. Table

The table is for placing optical devices. We can adjust the levelling of the table by three screws.

4. Reading Scales

The reading scales are for reading the angle value. It consists of a reading dial and two verniers which are placed symmetrically 180° apart along the edge of the dial. Two verniers can eliminate the systematic error caused by assembly. Its reading principle is the same as that of a vernier caliper.

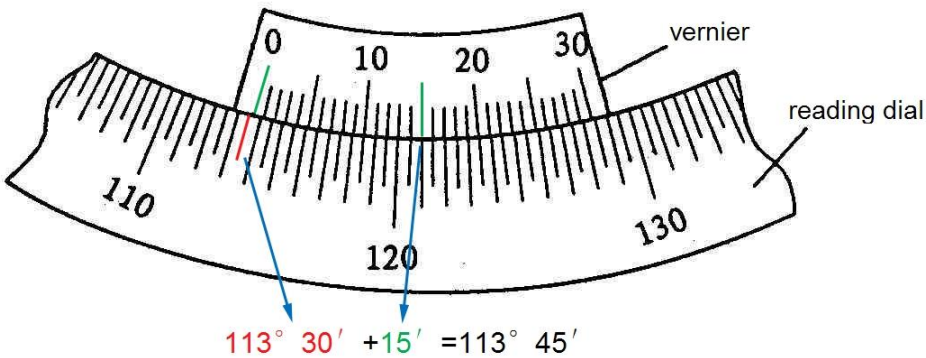


Fig. 5 Reading principle

$$\theta = \alpha_1 + \alpha_2 = \frac{\varphi_A + \varphi_B}{2} = \frac{(a_2 - a_1) + (b_2 - b_1)}{2} \qquad (1)$$

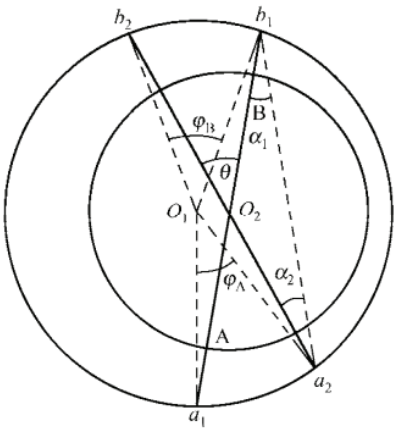


Fig. 6 The systematic error caused by assembly

Contents

1. Adjustment of the spectrometer

(1) Coarse adjustment

The purpose of coarse adjustment is to make the telescope, table, and collimator approximately horizontal.

Adjust the levelling screws of the collimator and the telescope to make their optical axes approximately horizontal.

Rotate the table, and observe whether the slot under the table is even. Adjust the screws a, b and c to make the table approximately horizontal.

Place the mirror on the table with the mirror facing the telescope. Rotate the table slowly and observe whether the cross image reflected from the mirror can be found in eyepiece view.

Rotate the table 180° to make the other side of the mirror face the telescope, and observe whether the cross image can be found in eyepiece view.

If the cross image can be observed in eyepiece view when both sides of the mirror are facing the telescope, it indicates that the coarse adjustment has been completed.

(2) Focus the telescope at infinity

If the cross image in eyepiece view is blurred, adjust the eyepiece focussing knob and objective lens focussing knob, so that the cross image can be seen clearly.

(3) Fine adjustment of the telescope and table

The purpose of fine adjustment is to make the cross image coincides with the upper horizontal line in eyepiece view.

Place the mirror on the table with the base of the mirror parallel to the line joining screws b and c. Rotate the table and the cross image reflected from the mirror can be found in eyepiece view.

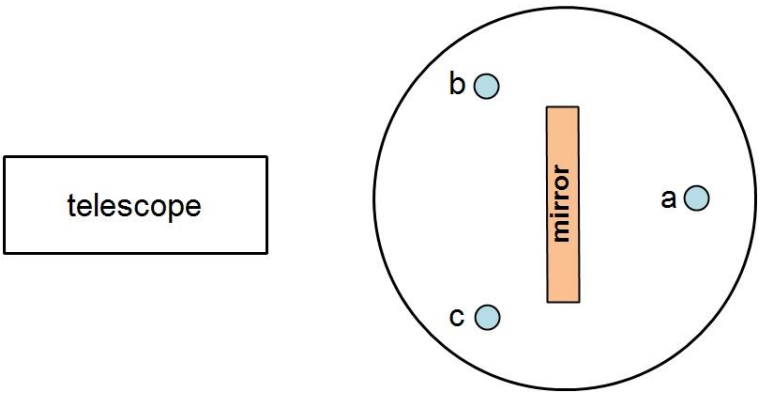


Fig. 7 Place the mirror for fine adjustment

Adjust the table screw a to move the cross image closer to the upper horizontal line by half (d/2), then adjust the levelling screw of the telescope to make the cross image coincides with the upper horizontal line.

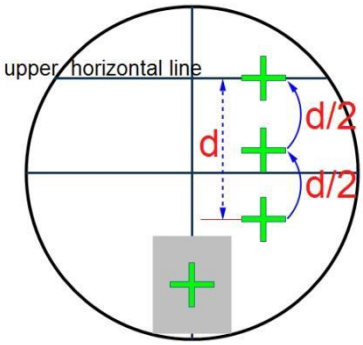


Fig. 8 Move the cross image

Rotate the table 180° so that the other side of the mirror faces the telescope, find the cross image in eyepiece view, and repeat the above operation.

Repeat the above operation several times until the cross image coincides with the upper horizontal line in eyepiece view when both sides of the mirror face the telescope.

Rotate the mirror 90° to make the base of the mirror perpendicular to the line joining screws b and c. Rotate the table and the cross image reflected from the mirror can be found in eyepiece view.

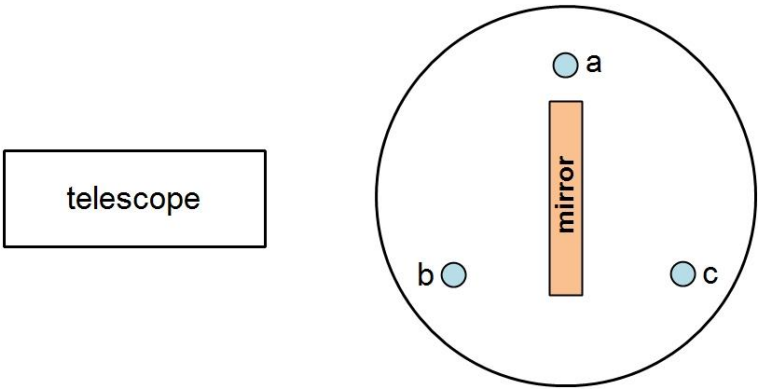
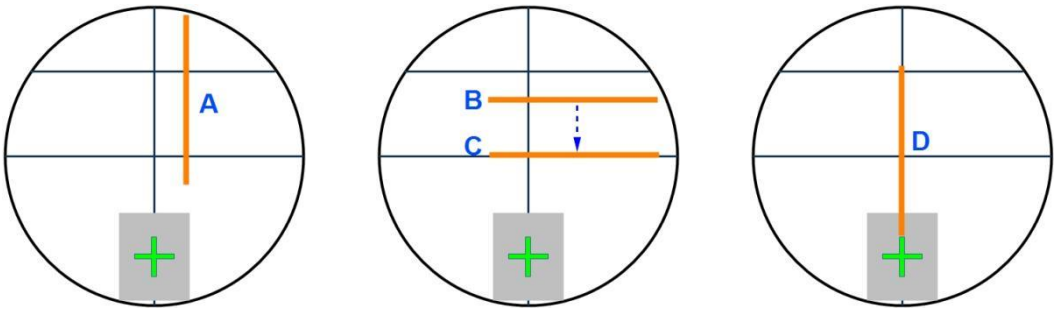


Fig. 9 Rotate the mirror 90°

Adjust the table screw b to move the cross image closer to the upper horizontal line by half (d/2), then adjust the table screw c to make the cross image coincides with the upper horizontal line. Rotate the table 180° and repeat the above operation. Repeat the above operation several times until the cross image coincides with the upper horizontal line in eyepiece view when both sides of the mirror face the telescope.

(4) Fine adjustment of the collimator



（1） （2） （3）

Fig. 10 Fine adjustment of collimator

Turn on the Sodium vapour lamp. Do not place a mirror on the table, rotate the telescope in line with the collimator. Find the image of the slit and adjust the width of the slit to be as thin as possible. Adjust the focussing knob to change the distance between the slit and the lens, so that the clear slit image can be seen from the telescope. Unlock the screw of the slit and rotate the slit by 90° to make the slit image horizontal. Adjust the levelling screw of the collimator to make the slit image coincides with the middle horizontal line. Rotate the slit 90° again to make the slit image vertical.

2. Use of the spectrometer

(1) Measure the vertex angle of a triangular prism by autocollimation

Lock the screws of table and vernier. Place the prism on the table, and rotate the telescope to face the two reflecting surfaces AB and AC (BC is the frosted surface). Find the reflected cross image, make the vertical line of the cross image coincides with the vertical line in eyepiece view. When the telescope faces plane AB, record the reading scales $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$. When the telescope faces plane AC, record the reading scales $\boldsymbol{\theta_1'}$ and $\boldsymbol{\theta_2'}$. The complementary angle of $\boldsymbol{\theta}$ is the vertex angle α of the prism.

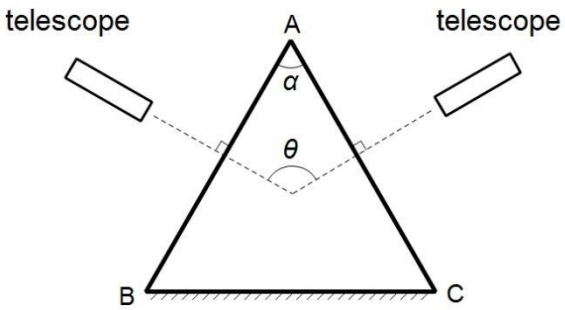


Fig. 11 Measuring the vertex angle of a triangular prism by autocollimation

$$\theta = \frac{1}{2}[|\theta_1' - \theta_1| + |\theta_2' - \theta_2|] \qquad (2)$$

$$\alpha = 180° - \theta = 180° - \frac{1}{2}[|\theta_1' - \theta_1| + |\theta_2' - \theta_2|] \qquad (3)$$

Unlock the screw of reading dial, and rotate the reading dial, repeat the above operation 6 times. α is averaged as the vertex angle of the prism.

(2) Measure the vertex angle of a triangular prism by reflection

Lock the screws of table and vernier. Place the prism on the table, with the frosted surface BC facing away from the collimator. The light beam emitted by the collimator is reflected on the two reflective surfaces AB and AC of the triangular prism. Rotate the telescope to observe the slit image reflected by the AB surface, when it coincides with the vertical line in eyepiece view, record the reading scales $\boldsymbol{\theta_3}$ and $\boldsymbol{\theta_4}$, then rotate the telescope to observe the slit image reflected by the AC surface, when it coincides with the vertical line in eyepiece view, record the reading scales $\boldsymbol{\theta_3'}$ and $\boldsymbol{\theta_4'}$. The vertex angle α of the prism is determined by Eq. (4).

$$\alpha = \frac{\beta}{2} = \frac{1}{4}[|\theta_3' - \theta_3| + |\theta_4' - \theta_4|] \qquad (4)$$

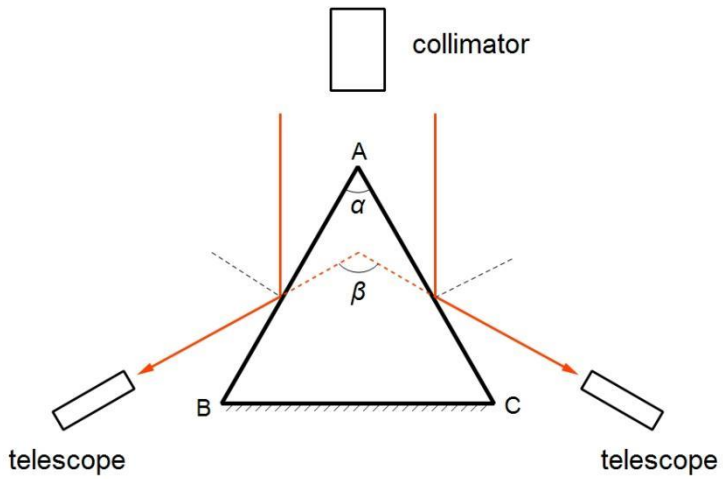


Fig. 12 Measuring the vertex angle of a triangular prism by reflection

Unlock the screw of reading dial, and rotate the reading dial, repeat the above operation 6 times. α is averaged as the vertex angle of the prism.

Precautions

1. In the process of adjustment, the previously adjusted parameters should not be adjusted again in the next step.
2. Do not touch the surface of optical devices with hands, do not break the mirror and prism.
3. Do not switch on and off the Sodium vapour lamp frequently.

Data Taking

Table1 Measure the vertex angle of a triangular prism by autocollimation

	θ_1	θ_2	θ_1'	θ_2'	α
1					
2					
3					
4					
5					
6					

Table2 Measure the vertex angle of a triangular prism by reflection

	θ_3	θ_4	θ_3'	θ_4'	α
1					
2					
3					
4					
5					
6					

Data Analysis

Calculate the uncertainty of the measured vertex angle of the triangular prism. The calculation steps are as follows:

Calculate the mean value of α.

Calculate the standard deviation of the mean value as the class A component of the uncertainty:

$$u_A = S_{\bar{\alpha}} = \frac{1}{\sqrt{n}}\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(\alpha-\bar{\alpha})^2}$$

Calculate the class B component of the uncertainty (Δ_m=1′=0.017°):

$$u_B = \frac{1}{\sqrt{3}}\Delta_m$$

Calculate the standard uncertainty of α:

$$u(\alpha) = \sqrt{u_A^2 + u_B^2}$$

Calculate the relative uncertainty of α:

$$\mathrm{E}(\alpha) = \frac{u(\alpha)}{\bar{\alpha}} \times 100\%$$

Questions

1. Why does the reading scales of a spectrometer use dual verniers?